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FULL PAPER



## Museum robot guides or conventional audio guides? An experimental study

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### ABSTRACT

In the last years, museums around the world began using museum guide robots as an alternative to audio guides. Thus, it is important to identify how these new museum guides can optimally interact with visitors and whether they are more efficient than conventional audio guides. In this paper, we present the results from an experimental study which compared conventional audio guides with robots with different 'personality'. These results demonstrate that people remember significantly more information when they are guided by a cheerful robot than when their guide is a serious one or a conventional audio system. Importantly, we also introduce the idea of two collaborative tour guide robots. This idea was inspired by cognitive studies showing that people remember more when they receive information from two different human speakers. Interestingly, our participants also liked more our collaborative robots, than any version of a single robot and/or audio system.

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acquisition

## 1. Introduction

Several museums around the world have started using museum guide robots as an alternative to audio guides and many researchers work on improving them either in terms of the interaction with visitors or in terms of their navigation within the museums [1]. Lindsey is an example of a long-term deployment of an autonomous tour guide robot [2]. However, it is not clear whether these tour guide robots perform better than conventional audio guides in general. Also, which presentation style by the robots is more effective. This paper aims to fill this gap in the literature. This study assesses the success of tour guide robot in terms of how much visitors remember from the tour and how enjoyable the tour was.

The most important findings of our study are that: (a) the emotional expression of the tour-guide robot(s) significantly improves the human knowledge acquisition; (b) the introduction of the new idea of a pair of collaborative robot guides, significantly improves the level of enjoyment of the visitors. (c) A cheerful robot is more efficient, in terms of knowledge gaining, than a conventional audio system, while two robots result in a higher level of enjoyment than the audio systems. Finally, based on our findings, it should be highlighted that certain style guide robots have been proved not to be more efficient than conventional audio systems.

## 2. Related work

Several studies [2,3] have demonstrated that humanoid robots tend to be more attractive to people than non-humanoid ones. However, in these studies participants did not see real robots, just avatars presented on a screen. This is rather critical as the human's preference for humanoid robots also stems from the fact that people can interact with them (e.g. [2]). For instance, people are better engaged in activities where the robots utilize gazes in order to highlight certain tasks/info. People also tend to pay more attention to robots that move their heads and look around [4]. Such interactive behaviour has also been shown to be efficient for museum guide robots [5]. Visitors remember more from a robot's storytelling when the robot looked at them [6,7].

The role of a robot's personality is also important. Goetz and his colleagues [3] compared two different types of robots, a cheerful and a serious one, which differ mainly in their speech styles. In their experiments there were two different tasks to be performed by the robots, one serious (physical exercise task) and one cheerful (a candy cooking task); participants interacted for longer with the cheerful robot in the cheerful condition and with the serious robot in the serious one. The authors considered museum guiding as a cheerful activity and therefore, they proposed that visitors in a museum will probably

prefer to spend more time with a robot that has a cheerful voice. However, since the candy activity is very different from a guided tour in a museum, it is not clear whether their results generalize to museum guide robots. In Takamasa's et al. [8] study the tour guide robot expressed a friendly attitude while it was greeting visitors by their names and the result is that 94.74% of the first time visitors expressed the desire to interact with the robot again. Finally, evidence from a comparison of a humanoid intelligent driving assistant with a smartphone device in a driving assisting task suggested that participants felt less stressed throughout the interaction with the robot and thus they enjoyed more the robot's services [9].

### 3. Present study

On one hand, the present study makes the common assumption that the purpose of museum (robot, audio, even human) guides is to convey information about the exhibits to the visitors. On the other hand, we also assume that these guides should make the visit to a museum enjoyable. Therefore, our experimental studies not only assess whether participants remember information about art exhibits (painting), but also whether they found this acquisition of knowledge enjoyable.

Guidance systems are available for most museums worldwide and there is a large variety in terms of hardware, software, content style, appearance etc. [10]. The museum visitors prefer the use of audio guides instead of other forms of interpretation and it is important to note that those who use them very rarely consult the information cards describing the artwork, or any of the other information aids at their disposal [11]. With the aid of audio guides, information such as title, artist's name, historical context, critical reviews, can be easily communicated to the visitors and the visitors tend to be accustomed to such guides. This is one of the main reasons why we are interested in comparing robot tour guides with existing audio systems: Is it really necessary to replace those systems with the much more expensive robots? And more importantly, do all the robots lead to better acquisition of knowledge and/or higher levels of enjoyment when compared with conventional audio systems?

Since we consider the acquisition of information and enjoyment of this acquisition critical, we assess the success of the robots in three ways. We established whether participants found the 'tour' enjoyable by using a custom-made adaptation of aesthetic valence questionnaire [12] at the end of the study. Second, we tested participants' knowledge acquisition with a multiple-choice memory test. Third, we also evaluated the strength of the memory instilled by the museum robot guide. This evaluation was based on a well-established psychology theory on

memory [15,16,17]: the stronger an item is memorized the faster participant respond when asked about this item. Hence, we measured participants reaction times in the memory test. This theory is based on the rehearsal processes in the memory. When an information enters the memory and is stored, mental representations are created through repetition. When this information needs to be retrieved, the faster the retrieval is done the more efficiently the storage has been done [13].

Given the research results described in the last section [3,8], it is clear that the personality of the museum guide robot can affect the information that people remember and that a cheerful personality robot can perform better as a tour guide in comparison with a serious personality robot (with respect to our three performance measures: more enjoyable, better memory and faster responses). And since the tone of conventional audio guidance is similar to serious robots, we expect the cheerful robot also to perform better than a conventional audio guide.

Furthermore, psychological experiments have shown that when two different human-speakers talk about the same topic, listeners can recall more information about the topic than in the case of a single speaker [14,15]. Hayashi and his colleagues [16] replicated a Japanese stand-up comedy with two collaborative robots versus two humans without testing participant's knowledge acquisition and found robots similarly preferable as comedians with humans. The present study investigated whether the same benefit can be also evidenced in robotic museum tour guides. Again this was done using the three performance measures: aesthetic valence questionnaire, accuracy and reaction times in a memory test. Our goal is to find the more efficient technology (robot or audio) for museum guides with respect to the previous work that was mentioned above.

## 4. Experimental studies

### 4.1. General method

#### 4.1.1. Stimuli

For the robots, we used P3 DX robots (Figure 1) and each one was connected to a tablet running Android which acted as its face (Figure 2). All the avatars for the robots' faces were based on women, as suggested by Goetz's [3] experimental results, where participants were shown different robot avatars (machinery, humanoid, women, men) in a screen and they choose the humanoid woman avatar as the more appropriate for a robot museum guide.

The robots' voice as well as the faces and the face expressions were created by Reallusion Crazy Talk 8. In order to compare cheerfulness and seriousness, we manipulated the robots in terms of their facial expression,



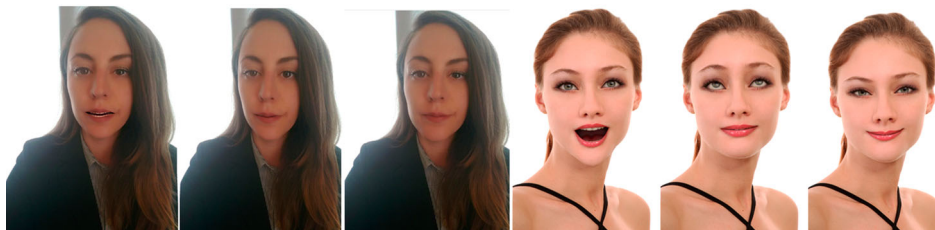
**Figure 1.** The robot design.

clothing style, voice and storytelling style (see Figure 2). As for the facial expressions, the cheerful robot made funny expressions and smiled for most of the time, while the serious robot had a professional look without any distinct facial expressions or smiles. Regarding the voice of the two robots, every robot had different voice, tone and speed, both of which were kept unchanged during the complete storytelling. Each of the robots was standing on one side of the painting they were talking about (Figure 3; left).

The videos of the avatars were evaluated by 10 students who participated in the pilot study for the questionnaire evaluation (see below). They all suggested that they reflect cheerfulness and seriousness as assumed by the experimenter.

#### 4.1.2. Storytelling

In general, the term ‘storytelling’ refers to telling a story about an object or event which may be slightly embellished. In our work storytelling refers to the talk about paintings given by a conventional audio guide and tour guide robot(s). The content of the robots’ storytelling referred to eight modern art paintings which appeared one by one on a Full HD large screen (see Figure 1). Each story lasted for seven and a half minutes and contained vital information about each painting. The differences in the story (i.e. our experimental manipulations) were that in the cheerful script the robot was excited about what it was saying regarding the artwork, showed more interest in its job and make some jokes and personal comments.



**Figure 2.** Serious avatar face in the left part. Cheerful avatar face in the right part.



**Figure 3.** Image with two robots on the left side. Image with one robot on the right side.

In the serious one, the robot was fully professional, without expressing any personal opinion or showing any positive feelings. For the two robots' condition we used the cheerful script. The two robots were talking in turns and each robot shared different content with the participants. This is important for the participants to have a clear categorization about the content of the speech of each robot. The 'tour' lasted 20 s longer because of the longer welcome and goodbye scripts of the two robots. The audio storytelling had the same content and duration with that of the serious robot condition.

#### 4.1.3. Procedure

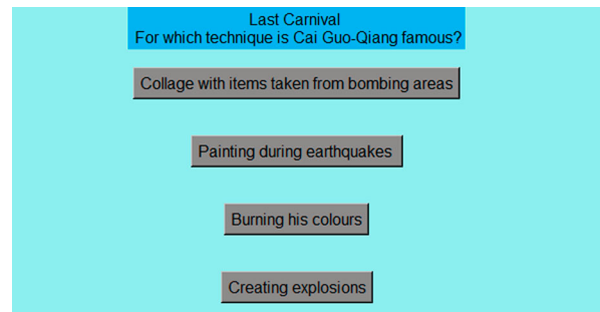
Overall the procedure mimicked a museum visit. The experimenter explained to the participants that tour guide robot(s) is/are going to give them information about 8 modern art paintings and that at the end of the 'tour', when they have seen and heard about all paintings they will answer questions on a computer regarding (a) the 'tour', (b) their experience with the robot and (c) some demographic characteristics. Then the experimenter introduced the robot and left. The robot welcomed the visitor, introduced itself, (following the recommendations by Rainer and Galán [17]) and asked the participant politely to follow it into the art exhibition which was at the other end of the room. When they arrived – robot and participant – at the screen presenting the paintings, the robot stood between the artwork screen and the participant. Each robot periodically altered its gaze looking at the participant, the artwork or the collaborative robot (if there). As seen in Figures 1 and 3 there was a specific space where the participant can stand. The whole procedure from the first welcome sentence to the final goodbye took 9 min.

After the participants entered the room and followed the procedure as mentioned above, the experimenter led them to a computer (in the same room close to the museum-set up) in order to answer the questionnaires (memory, aesthetic valence, demographics).

#### 4.1.4. Design

The experimental design is *between-participants* (i.e. in each condition we have different participants) and the robot experiments utilize the 'Wizard of Oz' approach [18].

**4.1.4.1. Memory test and graphical interface.** In order to eliminate the serial position effect (i.e. participants tend to remember more about the first and the final items of a list [18]) the memory test examined participants' memory for six of the eight paintings, excluding the first and the last painting. There was a total of 34 Multiple Choice Questions (MCQ) with only one correct answer



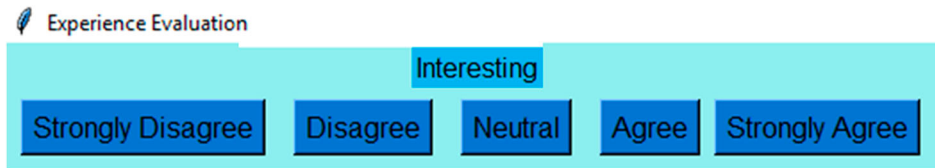
**Figure 4.** Example of a memory quiz question as appeared in the screen.

per question, about the style and the meaning of the paintings and the biography of the artist, as shown in Figure 4.

To identify the best questions, based on the criteria listed by Walker [19], we performed an initial pilot study. We created 80 MCQs and tested 100 people. After we analysed the MCQ, we decided to use the questions that had a response rate in the range of 35–80%; knowing that poor response rates can lead into biased conclusions [20] resulting in 34 questions. This way we choose the more appropriate questions based on the 'quality' of the question and not on the participants' prior knowledge, memory abilities etc.

Apart from the number of correct answers, we also measured the reaction times (RT) of the participants. For the questionnaires' graphical interface, we based our design on Dzulkifli [21] colour studies [21], who claimed that the background colour in which an information appears can affect peoples' memory and thus we used colours that inflate people's attention without affecting their memory.

**4.1.4.2. Aesthetic valence questionnaire (AVQ).** We created a Likert scale aesthetic valence questionnaire to measure the enjoyment of the tour. The participants instructed to answer, 'How they evaluate their experience from the tour' with 35 questions that appeared one by one in the screen in a scale from 'Strongly Disagree' (1) to 'Strongly Agree' (5) as shown in Figure 5. They can choose only one statement that is closer to what they feel. For the questionnaire's design and items, we used the recommendations of Marković's [12] work who evaluated the judgments of people when observing paintings. We used general words from [12] questionnaire, such as 'interesting', 'charming', 'hateful', 'coherent' – those referred to the participants general opinions – and in order to also ensure that we capture if participants were under the Uncanny Valley effect [22], we added also words such as 'creepy', 'ugly' and 'disgusting'



**Figure 5.** Example of the aesthetic valence questionnaire.

to the questionnaire. Our questionnaire gave us an overall aesthetic score – level of participants' enjoyment of the interaction – for each participant.

**4.1.4.3. Demographic and familiarity questions.** We created a questionnaire in which (a) we asked for the demographic characteristics of the participants (i.e. gender, age and academic level) and (b) we had eight Likert scale questions regarding the participant's relationship with modern art, museums and robots e.g. 'how familiar are you with modern art'.

#### 4.1.5. Data analysis

We analysed the questions from the MCQ with the Mann Whitney test [23] for the correct answers. We also analysed the reaction times with a Tukey's HSD multiple comparison test in order to reduce the probability of family error. In order to process the results of the AVQ we calculate a single number for each question by summing the answers of all the participants for this question, we applied a Tukey's HSD multiple comparison test in order to find the significant differences between the different conditions from the Aesthetic Valence Responses.

We also complete a correlation analysis, using SPSS, on the demographics and familiarity questions, memory test (both correctness of response and reactions) and the aesthetic valence answers. In general, knowing if the participants' familiarity level with our topic (robots, museums, modern art) are correlated with their scores in the given questionnaires after our manipulations and hence compatible, is useful for understanding variance structures in our data. Our hypothesis is that participants with prior experience with robots will enjoy more the robot condition, while those who are familiar with museums and modern art and, thus with conventional audio systems, are familiar that the audio condition more. Our hypothesis is based on Epstein [24], who support that familiarity is an important learning factor and thus we expect from the more familiar participants to have also higher learning acquisition scores in the MCQ. Finally, based on Goetz [3] participants who will consider the tour task as a serious job, would prefer (have higher scores in the AVQ) in the serious robot condition, while those who find it as a cheerful activity, would prefer a cheerful tour guide.

## 4.2. Experiment 1: audio versus serious tour guide robot

In Experiment 1 we compared the serious personal-ity tour guide robot with a conventional audio system (Figure 2 and accompanying video) on knowledge acquisition and the level of enjoyment of the visitors in an art exhibition. The purpose of Experiment 1 is to design a conventional audio guide condition and stress it against one serious robot guide. This comparison will let us know if we really need to use robot as guiding tour and if they perform the task better than the audio guides.

### 4.2.1. Method

**4.2.1.1. Participants.** The experiment was carried out in the robotics lab at the University of Birmingham. The total number of participants was 22, 11 in audio guide condition and 11 in the serious museum guide robot, 12 Males and 10 Females, aged 18–31 years old. 70% of them were MSc students, 25% BSc students and 5% PhD candidates and teaching staff. They all had normal or corrected to normal vision and hearing and proficiency knowledge of the English language. They were all recruited via official university email lists and they were paid for their participation.

**4.2.1.2. Procedure.** For the welcome state in the audio condition, the experimenter asked each participant to follow her into the exhibition part. When they arrived in front of the paintings, the experimenter started the audio guide and left. For the robot condition, the procedure was the same with the one described in the General Method. In the audio guide condition, participants heard the sound from a simple audio system which was placed on the side of the painting (i.e. similar position with the robots in order to keep the source of the sound unchanged for all the experiments).

### 4.2.2. Results

There was no significant difference ( $p = .46$ ) between the AVQ score in the serious robot condition ( $M = 39.1$ ,  $SD = 4.79$ ) and the audio guide condition ( $M = 37.3$ ,  $SD = 5.72$ ). There was a difference between MCQ score in the audio condition ( $M = 15.36$ ,  $SD = 3.58$ ) and the serious robot ( $M = 12.6$ ,  $SD = 4$ ), as shown in Figure 9.

However, a Mann–Whitney  $U$ -test indicated that this difference was not significant ( $U = 29.5, p = .078, r = .53$ ). Moreover, the difference between reaction times in the audio guide ( $M = 9$  s,  $SD = 2.37$ ) and the serious robot ( $M = 8.14$  s,  $SD = 3$ ) was not significant as indicated by the Tukey test ( $p = .88$ ).

Additionally, the correlation analysis shows no significant relationship between the demographic characteristics, the gender ( $r(10) = .236, p = .486$  for audio and  $r(10) = .058, p = .86$  for serious), the age ( $r(10) = -.49, p = .885$  for audio and  $r(10) = -.141, p = .68$ ) and the academic level ( $r(10) = .131, p = .70$  for audio and  $r(10) = .369, p = .26$ ) played no role in their memory test scores.

Furthermore, for the familiarity questions in the audio condition, we found a positive correlation with the AVQ score and their enjoyment of museum visits in general ( $r(10) = .71, p = .014$ ), appreciation of modern art ( $r(10) = .838, p = .001$ ) and how familiar they were with modern art ( $r(10) = .785, p = .04$ ). As for the correct answers and reaction times we found no correlation with the familiarity questions. In the serious robot condition we found a positive correlation between the MCQ scores and how familiar participants were with modern art,  $r(10) = .765, p = .006$ . For the AVQ there was no correlation with any of the familiarity questions.

#### 4.2.3. Discussion

There were no statistically significant differences between an audio guide and a museum guide robot in terms of participants' enjoyment, knowledge acquisition and memory quality. It seems that the aesthetic outcome is higher in the audio condition for the participants who are visiting museums frequently and thus they are familiar with the conventional audio systems which they probably use frequently. Moreover, our assumption regarding the positive correlation between the MCQ score and participants' familiarity with modern art in the serious robot condition was that the visitors of a museum who are familiar with such museum's exhibition remember more from the storytelling and thus they pay more attention to the tour guide robot when the robot have a serious personality.

### 4.3. Experiment 2: audio versus cheerful tour guide robot

Based on the research results described in the introduction section, the personality of the tour guide robot can strongly affect the information that people remember and the related knowledge they gain from the robot's storytelling, but this was not tested with an actual museum robot.

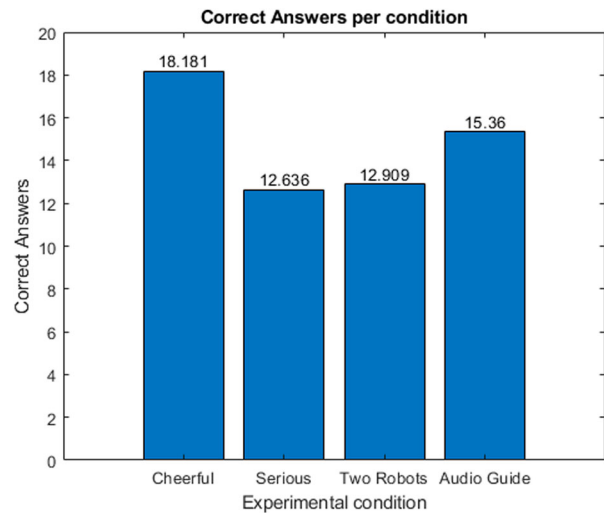


Figure 6. Average correct answers per condition.

#### 4.3.1. Participants

The total number of participants was 11, 6 Males and 5 Females with similar characteristics with Experiment 1.

#### 4.3.2. Results

The AVQ score in the cheerful robot condition ( $M = 38.6, SD = 5.96$ ) was 25% higher than in the audio condition, but as showed by the Tukey test that applied the difference was no significant ( $p = .703$ ). Additionally, participants in the cheerful robot condition had similar score with those in the serious robot condition,  $p = .98$ .

In terms of the MCQ score, in the cheerful-robot condition ( $M = 18, SD = 5.7$ ) the score was higher but not statistically significant than the audio condition based on the  $U$ -test ( $U = 49.5, p = .49, r = .2$ ) and statistically significant higher than the serious robot condition ( $U = 24.5, p = .020, r = .703$ ) as shown in Figure 6. Tukey's Post Hoc test results confirmed the difference between cheerful and serious robot condition at  $p = .022$ . The reaction time in the cheerful robot guide ( $M = 8.6$  s,  $SD = 3$ ) as indicated by the Tukey test we applied, was no significant when comparing with the serious guide ( $p = .98$ ) and the audio guide ( $p = .99$ ), as shown in Figure 8.

Additionally, correlation between the demographic characteristics and the MCQ scores showed that the gender ( $r(10) = .394, p = .231$ ), the age ( $r(10) = -.584, p = .086$ ) and the academic level ( $r(10) = .281, p = .402$ ) played no role in the memory (Figure 7).

For the familiarity questions in the cheerful condition, we found a positive correlation between the participants correct answers in the MCQ and if they like visiting museums ( $r(10) = .651, p = .03$ ). As for the reaction time, we found a positive correlation on how familiar



**Figure 7.** The two collaborative robots' avatars.

they were with robots ( $r(10) = .702$ ,  $p = .016$ ) and a negative correlation on their appreciation of modern art ( $r(10) = -.580$ ,  $p = .061$ ). Moreover, there is a positive correlation between high aesthetic scores and how much they appreciate modern art in the cheerful condition ( $r(10) = .637$ ,  $p = .035$ ).

#### 4.3.3. Discussion

The outcome of this analysis is that the visitors of a museum remember more from the cheerful robot storytelling, and thus they gain more knowledge from a museum robot with a cheerful personality. On the other hand, their level of enjoyment (AVQ) is similar regardless of robot's personality and style. Furthermore, visitors of a museum who like visiting museums pay more attention to the cheerful personality robot. Moreover, participants who are familiar with robots have quicker response times and thus pay more attention to the cheerful robot, while they have slower reaction times and thus pay less attention when they appreciate modern art. On the other hand, the aesthetic outcome is higher in the cheerful robot condition for the participants who appreciate modern art ( $r(10) = .637$ ,  $p = .035$ ).

#### 4.4. Experiment 3: two collaborative museum guide robots versus single robot and audio

The purpose of the current experiment is to examine if two collaborative cheerful robots are better tour guides than a single one and the audio system. We

integrated the robot into the optimal tour guide characteristics – based on the bibliography section – and the results of Experiment 2. Based on research results when two humans perform a storytelling [15], we expect that visitors will remember more when they have two guide robots than in the case, they have a single robot guide. Additionally, since participants had higher scores in Experiment 2 in the one cheerful robot condition in comparison with both serious robot and audio guide, we expect the two collaborative robots to outperform all three previous conditions. The Methodology and Data Analysis was the same that described in the General Methodology Section.

##### 4.4.1. Participants

The total number of participants was 11, 5 Males and 6 Females with similar characteristics with Experiment 1.

##### 4.4.2. Results

The AVQ score in the two collaborative robots' condition ( $M = 41.7$ ,  $SD = 5.7$ ) was statistically significantly higher than the audio condition as indicated by the Tukey test results ( $p = .004$ ). The difference between the two collaborative robots and the cheerful one was borderline significant a ( $p = .051$ ) while the difference between the two collaborative robots and the serious one was not confirmed ( $p = .064$ ).

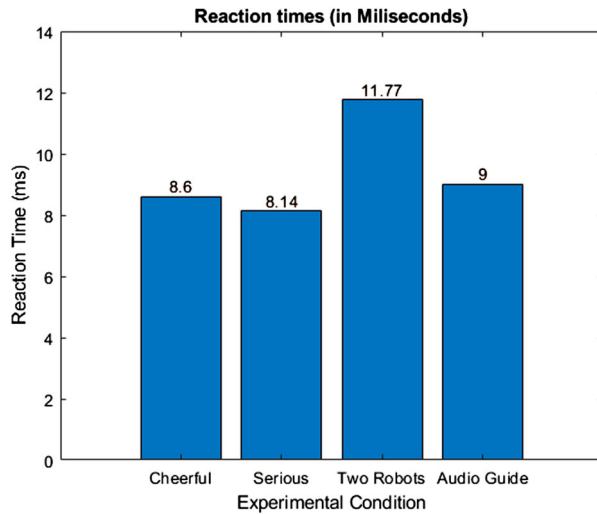
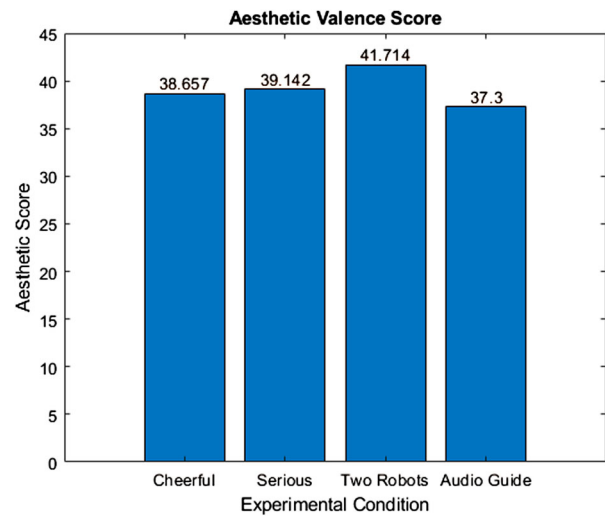
Regarding the MCQ scores, participants in the two-robots condition ( $M = 12.9$ ,  $SD = 3.5$ ) scored similarly with the serious robot condition ( $M = 12.63$ ,  $SD = 4$ ) (see Figure 6). However, in the cheerful single-robot condition, the correct answers of the participants were 30% more than in the two-robot condition and this difference is significant as demonstrated by the two tail  $U$ -test ( $U = 24$ ,  $p = .018$ ,  $r = -.712$ ) and Tukey test ( $p = .032$ ), also shown in Table 1. The MCQ score in the audio condition was 21% higher than the two-robots condition, but the difference proven not significant based on the  $U$ -test that performed ( $U = 32$ ,  $p = .114$ ,  $r = -.47$ ).

Participants' reaction times in the two robot condition ( $M = 11.77$  s,  $SD = 2.846$ ), proven to be significantly slower as indicated by the Tukey test than in (a) the cheerful single-robot condition ( $p = .045$ ), and (b) the serious robot one ( $p = .017$ ). On the other hand, the difference with the audio guide condition, as demonstrated by the Tukey test was no significant ( $p = .089$ ) and shown in Figure 8.

The correlation between the demographic characteristics and the participants scores in MCQ, AVQ and reaction times demonstrated no correlations, just as in Experiment 1. Similarly, no correlation found between those scores and all the familiarity-related questions.

**Table 1.** The  $p$ -values of the comparison of all conditions.

|                  | $p$ Value of average<br>cor. answers | $p$ Value of average<br>reaction time | $p$ Value of average<br>aesthetic scores |
|------------------|--------------------------------------|---------------------------------------|------------------------------------------|
| Cheerful-Serious | .02*                                 | .98                                   | .98                                      |
| Serious-2Robots  | .49                                  | .017*                                 | .064                                     |
| Cheerful-2Robots | .018*                                | .045*                                 | .051*                                    |
| Cheerful-Audio   | .49                                  | .99                                   | .7                                       |
| Serious-Audio    | .078                                 | .88                                   | .46                                      |
| 2Robots-Audio    | .114                                 | .098                                  | .004*                                    |

**Figure 8.** Average reaction time per condition.**Figure 9.** Average aesthetic valence score.

#### 4.4.3. Discussion

Shorter response time means better memory encode and so participants seem to gain more knowledge from the single robot (either cheerful or serious) than to the collaborative ones.

With respect to the number of correct answers in the memory test, the audio systems stands in the middle, between the cheerful robot condition where participants had significantly more correct answers and the serious and the 2-robots conditions where participants had less correct answers (see Figure 6). On the other hand, participants tend to like much more the robot conditions than the audio one as indicated by the highest score in the aesthetic valence questionnaire (see Figure 9). The statistically significant difference did not confirmed from the Tukey test for the reaction time between the two robots and the audio condition and for the AVQ scores between the two robots and serious one was close to significant at  $p = .064$ . This effect indicates that with more participant it would have been likely to be significant.

### 5. General discussion and conclusions

Our study focused on the interaction between tour guide robot(s) and museums' visitors. We were interested mainly in how the robot(s) can improve the visitor's

knowledge about the artwork as well as in the aesthetic valence of the visitor's experience. Our study was initially motivated by the proposals of Goetz et al. [3]; they suggest that a cheerful personality robot would be better as a museum guide compared to a serious one. However, they have not tested their assumptions in any realistic setting. Following the proposals explicitly listed in [3], we have conducted a realistic experiment and we confirmed that a cheerful robot is more efficient than a serious one for museum guidance. Additionally, we investigated the novel idea that two guide robots may be better than one, since psychological experiments showed that people tend to remember more when there are two speakers providing information than in the case of a single speaker. Finally, we compared the performance of tour guide robot(s) with a conventional audio guide system in order to make sure that robots are more efficient as museum guides than the technologies they are about to replace.

The first outcome is that the participants retained less knowledge from the two-robot presentations than in the case of a single cheerful robot. In particular, a single cheerful robot is the clear winner in the memory test since the participants had more correct answers in

comparison with both the single serious, the collaborative robots' and the audio guide cases; thus, we believe that they paid more attention in the cheerful personality robot. Moreover, in terms of the reaction times, the participants interacting with the serious robot condition had similar scores with those guided by the cheerful robot and the audio guide; the two-robots led to higher reaction times. Our assumption behind the low memory scores and the high reaction times in the two robots' experiment is that the visitors paid more attention on how the robots interacted with each other and this distracted them from the storytelling. Hence overall, our results suggest that a single cheerful robot is better, for knowledge acquisition, when compared with two robots, while also the conventional audio guide can be better than a serious robot (in terms of gain knowledge).

On the other hand, our results clearly demonstrate that the two collaborative robots lead to a more pleasant experience than single robots (either cheerful or serious) and conventional audio guides. This has been identified by the aesthetic valence results; in the two robots' scenario the scores were higher than in any other condition.

In terms of participant's characteristics, based on the familiarity questions, we have also identified some interesting points. Based on our analysis, it seems that 'liking to visit museums' can be seen as an indication of an emotional relationship with museums which triggers a joyful feeling to the participants; thus, the cheerful robot is more suitable for them. On the other hand, participants that 'appreciate modern art' consider the museum visit as a 'serious' task, thus the serious robot is a more efficient guide for them. Moreover, it is worth mentioning that no such correlations were found in the two robots' case. As a result, we believe that the number of robots is equally important with the robot's personality and it can affect people's way of perception. Another explanation is that since, based on our findings the two robots trigger higher aesthetic valence, participants did not evaluate them with the same behavioural norms that they use in the evaluation of the personality of a single robot. Moreover, regarding the audio guide, participants who have previous experience with museums and thus they are familiar with audio guides, had higher scores in aesthetic questionnaires (i.e. tend to like them more). Finally, in some cases, i.e. the MSQ scores in the comparison between serious robot and audio condition, the effect size is close to statistical significance and we believe that by increasing the number of participants the effect size will be increased accordingly. This hypothesis needs further testing. Similarly, for the Tukey's test results in the comparison between the AVQ scores of the two robots and the serious one and in the comparison between the reaction time of the two robots and the audio condition

where the difference is barely significant and thus we expect that with a bigger participants' size most probably it will still be proven significant.

Overall, we believe that our results can also be applied beyond tour guides in, for example, the educational field, so as to enhance students' learning, memory and knowledge acquisition as well as to make the interaction with robots more pleasant. Additionally, we propose the use of two robots in entertainment tasks.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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